recall: solve $Ax = b$, A is a square matrix. TFAE

(a) $Ax = 0$ only has zero solution.

(b) $Ax = b$ has unique solution for some $b \in \mathbb{R}^n$.

(c) $Ax = b$ has unique solution for all $b \in \mathbb{R}^n$.

(d) $\operatorname{rank} A = n$.

(e) $\det(A) \neq 0$. Moore-Penrose

(f) A is invertible. inverse.

(g) A^{-1} exists.

matrix multiplication

can we compute matrix like real numbers.

e.g. $AB = BA$. × $A(BC) = (AB)C$ ✓

$AB = 0 \Rightarrow A = 0, B = 0$ ×.

$AB = AC, A \neq 0 \Rightarrow B = C$ ×. for square matrix $\Leftrightarrow \det(A) \neq 0$.

$$A=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix} \qquad B=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$$

act on the plane.

(or on a subspace. i.e. a line)

rotate by θ. reflect by x-axis.

Linear transformations. $AB \neq BA$.

把两个矩阵的乘法计算转变为两个映射的复合是否相同.

$(AB)C = A(BC)$ ✓.

旋转可以交换. 所旋转矩阵的乘法可以交换.

rank $A \in M_{m,n}(\mathbb{R})$.

e.g. $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ $\quad Ax = \begin{pmatrix} x_1 + x_3 \\ x_2 + x_3 \\ x_1 + x_2 + 2x_3 \end{pmatrix} = \begin{pmatrix} u \\ v \\ u+v \end{pmatrix}$

$\operatorname{im} A := \{(u, v, w) \in \mathbb{R}^3 : w = u + v\}$ a plane.

$\operatorname{rank} A = \dim \operatorname{im} A$. geometric meaning of rank.

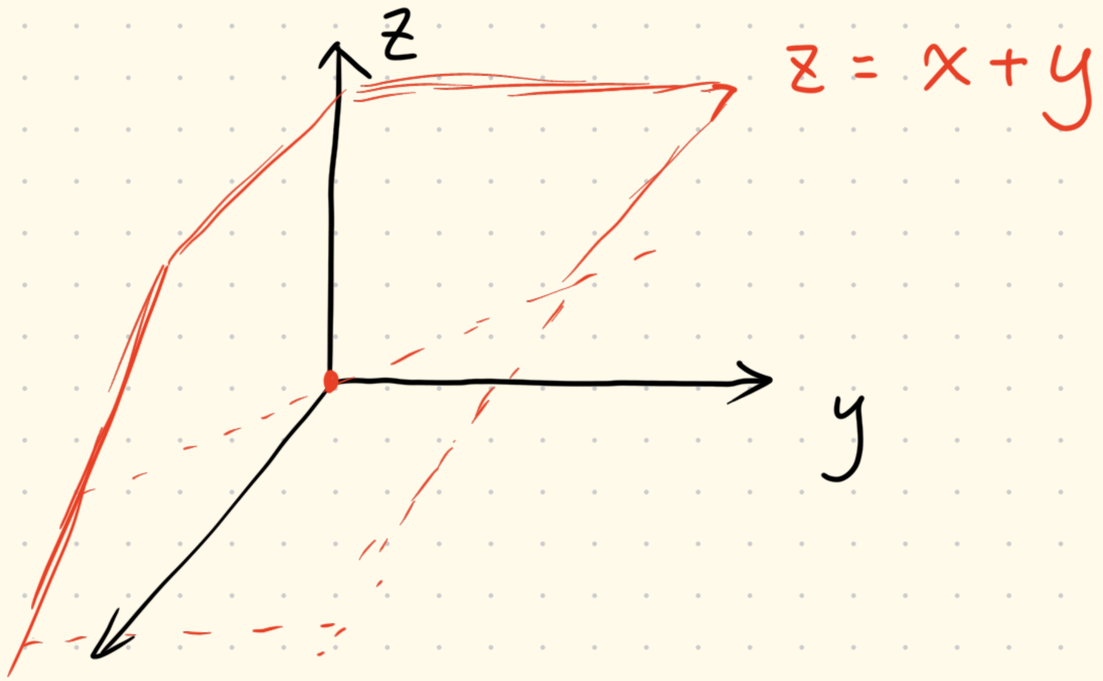
Moreover, $\ker A = \{(u, v, w) \in \mathbb{R}^3,\ v = u,\ w = -u\}$.

$\dim \ker A + \operatorname{rank} A = n$. (rank-nullity theorem).

$A^{rd} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. reduced form

$\operatorname{im} A := \{(u, v, w) \in \mathbb{R}^3 : w = 0\}$.

$\ker A := \{(u, v, w) \in \mathbb{R}^3 : x = y = -z\}$.



$z = x + y$



Theorem. $A, B \in M_{m,n}(\mathbb{R})$, $B = E_k \cdots E_1 A$ by finite many times elementary row transformation. Write $P := E_k \cdots E_1$. Then

(a) the row vectors of matrix A and B have the same linear relations.

(b) $\operatorname{im} B = P(\operatorname{im} A)$, $\ker B = \ker A$.

(c) $\operatorname{rank} A = \operatorname{rank} B$.

$$[b_1 | \cdots | b_n] = P[a_1 | \cdots | a_n] = [Pa_1 | \cdots | Pa_n].$$

proof. P is invertible.

$$\sum_{j=1}^{n} c_j b_j = 0 \iff P\left(\sum_{j=1}^{n} c_j a_j\right) = 0. \iff \sum_{j=1}^{n} c_j a_j = 0.$$

$\Longrightarrow \ker A = \operatorname{im} B$.

$$\operatorname{im} B = \{PAx : x \in \mathbb{R}^n\} = P\{Ax : x \in \mathbb{R}^n\} = P \operatorname{im} A.$$

$\dim \operatorname{im} B = \dim P \operatorname{im} A \Longrightarrow \operatorname{col\,rank} A = \operatorname{col\,rank} B$.

Intro: dimension finite ✓easy. infinite (possible shapes of ideal vibrating string).

think: $A \in M_{m\times n}(\mathbb{R})$. $P \in GL_m(\mathbb{R})$. $Q \in GL_n(\mathbb{R})$.

$B = PAQ$. $\operatorname{Im} B = P \operatorname{Im} A$, $\ker B = Q^{-1}(\ker A)$.

$\operatorname{rank} A = \operatorname{rank} B$.

$\operatorname{colrank} A = \operatorname{raw} \operatorname{rank} A$.

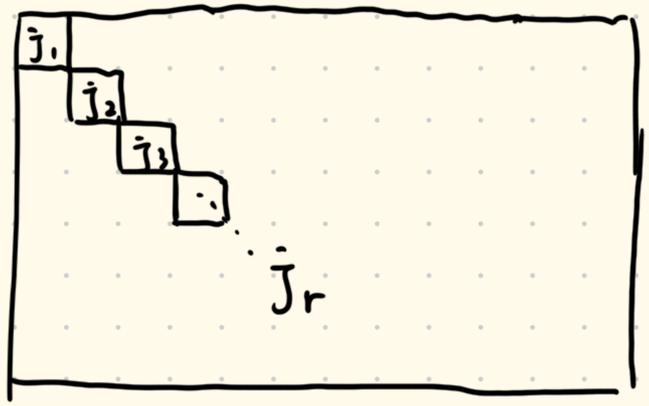
colrank: the number of maximal linear independent column vector, independent of choice (exist because of axiom of choice and uniqueness is easy to prove)

row rank A. the same.

Theorem. $A \in M_{m\times n}(\mathbb{R})$. row rank A = col rank A.

row rank A = row rank A^{rd}, col rank A = row rank A^{rd}.

A^{rd} is uniquely determined.



row rank A = r = col rank A.

column maximal linear independent set is $\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix}\ \begin{pmatrix}*\\1\\0\\\vdots\\0\end{pmatrix}\ \cdots\ \begin{pmatrix}**\\1\\0\\\vdots\\0\end{pmatrix}$.

rank normal form. $A \in M_{m,n}(\mathbb{R})$,

$\exists\, R \in GL_m(\mathbb{R})$, $C \in GL_n(\mathbb{R})$, s.t. $RAC = \begin{bmatrix} I_{r\times r} & 0 \\ 0 & 0 \end{bmatrix}$.

exercise. 2.4. 9.

A. B square matrices. If $I + AB$ is invertible. so does $I + BA$.

$(I + AB)x = 0$ only has zero solution.

Assume that $(I + BA)x = 0$. Then

$A(I + BA)x = 0$, $(A + ABA)x = 0$.

$\Rightarrow (I + AB)Ax = 0 \Rightarrow Ax = 0 \overset{\text{take back}}{\Rightarrow} Ix = 0 \Rightarrow x = 0$.

$\rightsquigarrow I + BA$ is invertible.

2.4 2(2) $A \longmapsto AQ$, $I \longmapsto IQ = Q$. where

$Q := E_1 \cdots E_k$. $AQ = I \Rightarrow Q = A^{-1}$.